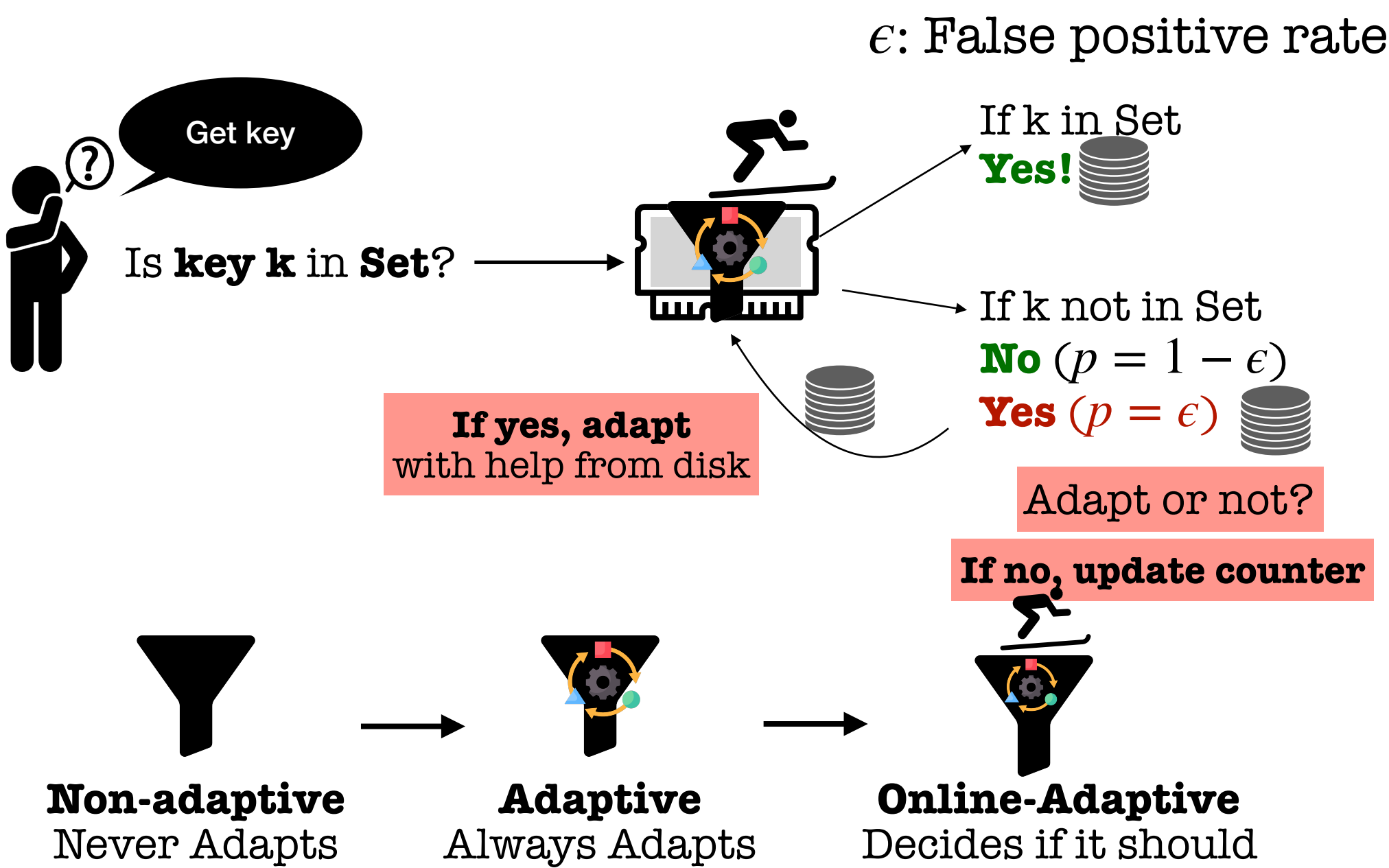
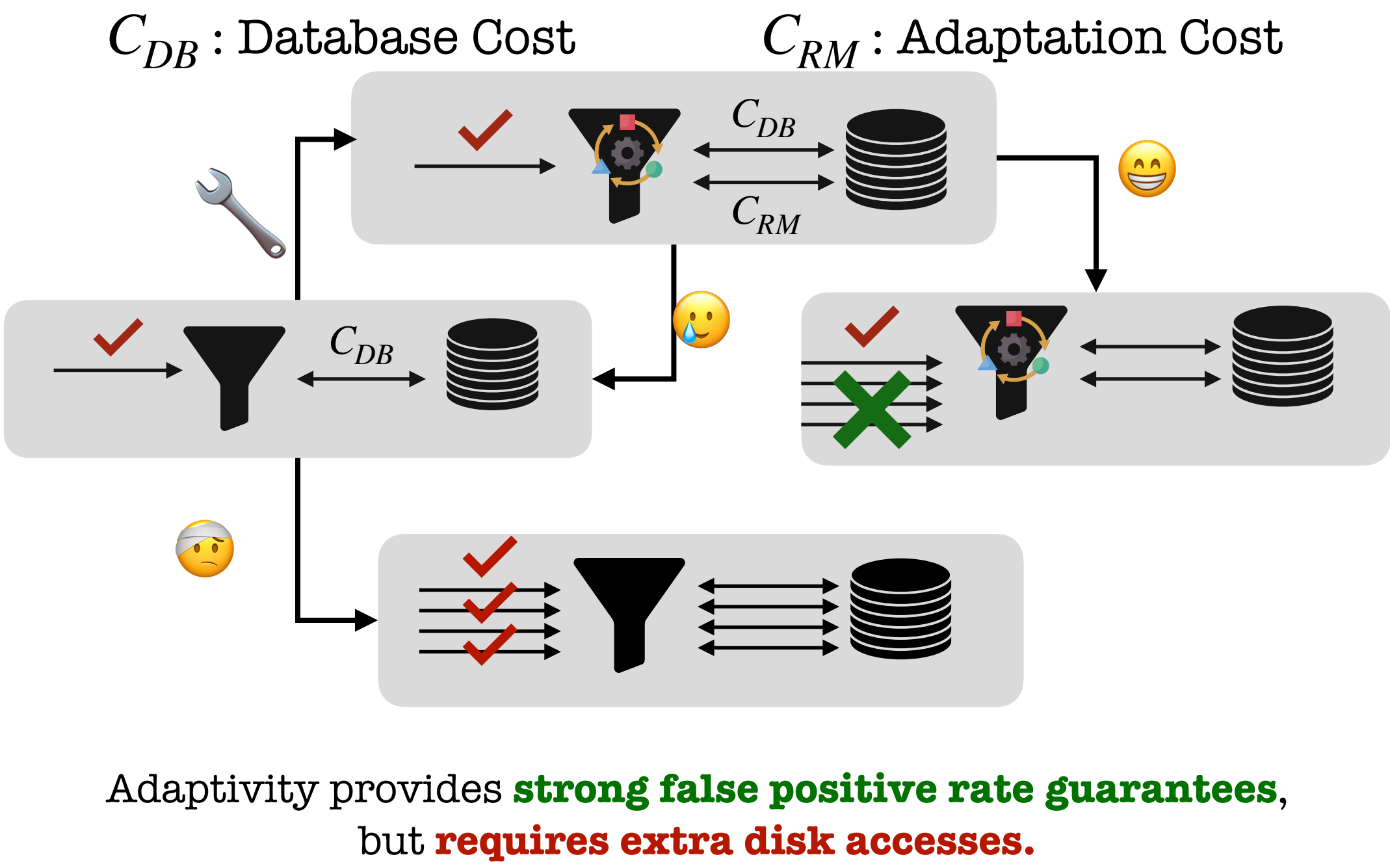


The Ski-Adaptive Quotient Filter



Why online adaptivity?



To Adapt or Not?

Ski Rental Algorithm

Rent until day $\lfloor B/r \rfloor$,
buy next day.

$\lfloor B/r \rfloor$: Break-even day

B : Buying cost
 r : renting cost

$$B = C_{DB} + C_{RM}$$

$$r = C_{DB}$$

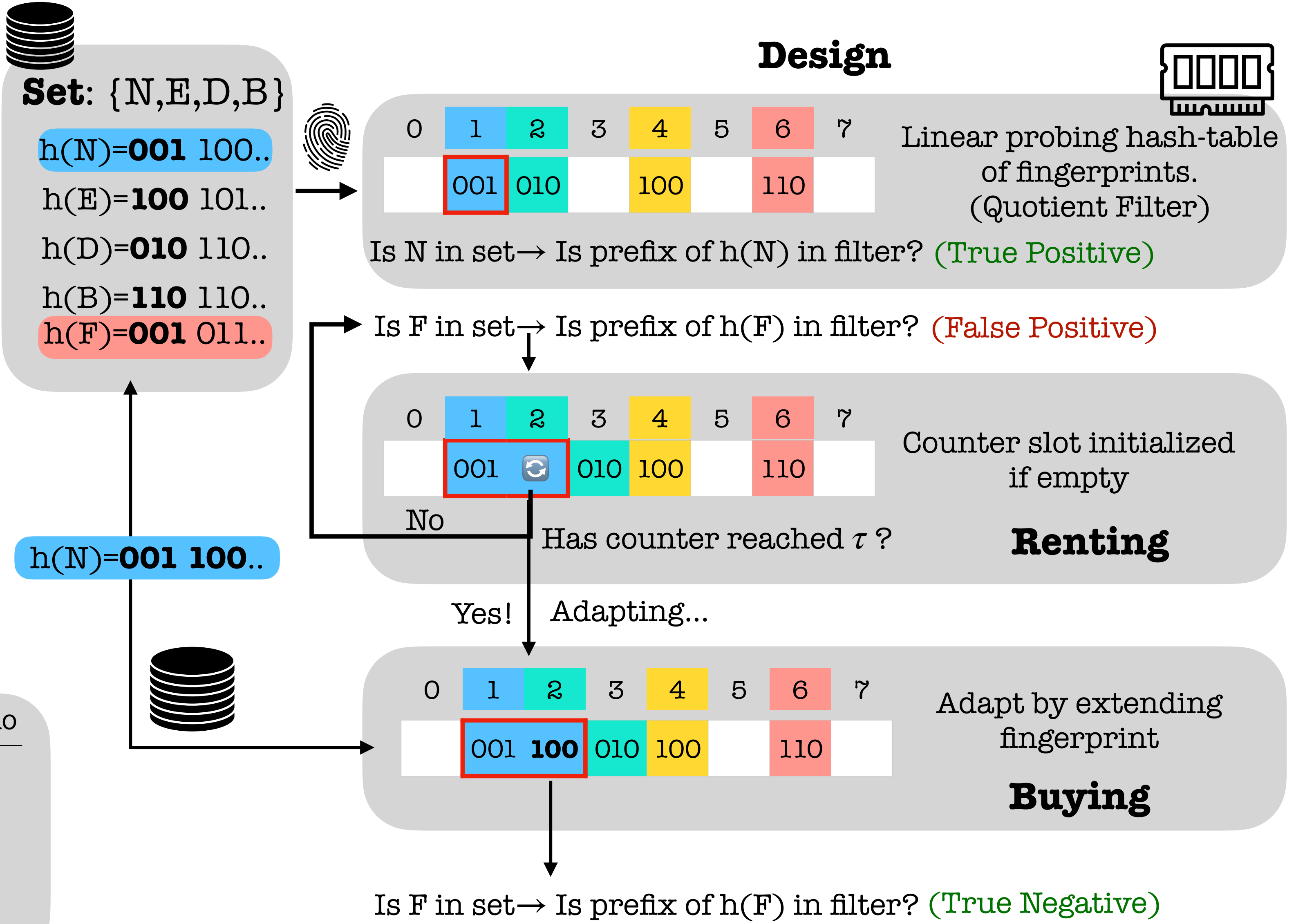
$$\alpha = C_{RM}/C_{DB}$$

$$\lfloor B/r \rfloor = 1 + \lfloor \alpha \rfloor$$

τ : False positive repetitions.
(Adapt on $\tau + 1$)

α	τ	Comp. Ratio
≥ 1	$\lfloor \alpha \rfloor$	≤ 2
$[0.618, 1)$	1	$\frac{(2 + \alpha)}{(1 + \alpha)}$
$[0, 0.618)$	0	$(1 + \alpha)$

$\alpha \approx 1$ is most representative setup



Experimental evaluation

Throughput (Higher is better)

Adaptive filter mostly performs best.
The SkiQF is 1.5-competitive.

